

Lessons 7 Biconditionals; Proof by Contrapositive

1. Prove or disprove:

For all integers x and y , x and y are even if and only if $x + y$ is even.

Prove the following biconditionals:

2. An integer x is even if and only if $x/2$ is an integer.
3. An integer x is odd if and only if $-x$ is odd.
4. An integer is odd if and only if it is a sum of two consecutive integers.
5. Let a be an integer. Then $a^2|a$ if and only if $a = 0, -1$, or 1 .
6. Prove the following proposition:
Let $x \in \mathbb{Z}$ and p, q be distinct primes. Then $p|x$ and $q|x$ if and only if $pq|x$.
You may use the following theorem:
For every prime p and all integers a, b , if $p|ab$, then $p|a$ or $p|b$.
7. Prove the following statement:
Let x be an integer and p a prime. Then $p|x$ if and only if $p|x^2$.
You may use the following theorem:
For every prime p and all integers a, b , if $p|ab$, then $p|a$ or $p|b$.

Prove by contrapositive:

8. If $a - b$ is odd, then a and b have different parities.
9. For every integer y , if y^2 is odd, then y is odd.
10. For every integer y , if y^3 is even, then y is even.
11. If the product of two integers is even, then at least one of the factors is even.
12. If the product of two integers is odd, then the integers are not consecutive.
13. Let n be an integer. If $4 \nmid n^2$, then n is odd.
14. For every integer y , y is even if and only if $4|(y^2 + 2y)$.
Note: Prove one direction directly.
15. For all nonzero real numbers x and y , $|x + y| = |x| + |y|$ if and only if x and y have the same sign.
Note: Prove one direction directly.
16. Let $x, y \in \mathbb{R}$. If xy is irrational, then x is irrational or y is irrational.
17. Let x be a real number. If x is irrational, then $-x$ is irrational.